Kunal Chopra

Embedded Firmware Engineer

RESUME_1437

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Total Experience: 2 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 2 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: C++, UART, SPI, Git, I2C, FreeRTOS, Oscilloscope, STM32, C, JTAG, ARM Cortex-M, Embedded Linux

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | BrioTech Labs | Jul 2024 - Present

- Improved battery runtime by 39% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 33% using interrupt priority and queue tuning
- Validated 806 hardware test vectors across protocol and timing corner cases

PROJECTS

Motor Control Interface | Git, I2C, ARM Cortex-M, JTAG

- Implemented CAN message handling, fault logging and deterministic state transitions.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Bootloader Update Utility | ARM Cortex-M, I2C, FreeRTOS, JTAG

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Sahyadri College of Engineering, 2024
Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

ARM Cortex-M Workshop (2026)